NUMERICAL METHODS FOR PDES
Lecture 7

For

$$\mathcal{L}u = -\nabla\cdot(A\nabla u) + \underline{b}\cdot\nabla u + cu$$

$F \in V'$, $\Omega \in \mathbb{R}^d$

$$a(u,v) = \int_\Omega A\nabla u\cdot\nabla v + \int_\Omega \underline{b}\cdot\nabla u\, v + \int_\Omega cuv$$

↑

If $\underline{b} \in W^{1,\infty}(\Omega)$ and

$$c - \frac{1}{2}\nabla\cdot\underline{b} > 0 \qquad \text{(suff. cond)}$$

Then the weak problem: Find $u \in V = H_0^1(\Omega)$:

$$a(u,v) = F(v) \quad \forall v \in V$$

is well-posed by L.M.

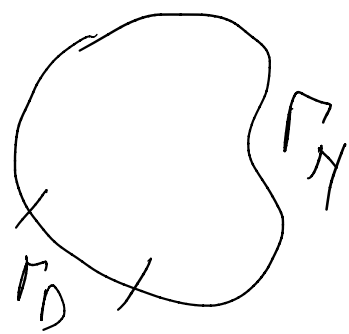
Mixed (Dirichlet-Neumann) b.c.

$$\begin{cases} \mathcal{L}u = f \in L^2 & \text{in } \Omega \\ u = g_D & \text{on } \Gamma_D \quad (= \text{Dirichlet boundary}) \\ A\nabla u \cdot n = g_N & \text{on } \Gamma_N \quad (= \text{Neumann boundary}) \end{cases}$$

$$\partial\Omega = \overline{\Gamma}_D \cup \overline{\Gamma}_N \; ; \; \Gamma_D \cap \Gamma_N = \emptyset$$

$$|\Gamma_D| \neq 0$$

↑ measure (lenght, area)



From $\mathcal{L}u\, v = f v$

$$\int_\Omega A\nabla u \cdot \nabla v - \int_{\partial\Omega = \overline{\Gamma}_D \cup \overline{\Gamma}_N} A\nabla u \cdot n\, v + \int_\Omega \underline{b} \cdot \nabla u\, v + \int_\Omega c u v = \int_\Omega f v$$

Choose or trial and test spaces

$$V_{g_D} = \{ v \in H^1(\Omega) : v|_{\Gamma_D} = g_D \}$$

(Dirichlet cond imposed on space – "essential" b.c.)

$$V_0 = \{ v \in H^1(\Omega) : v|_{\Gamma_D} = 0 \} = H^1_{\Gamma_D}(\Omega)$$

$$-\int_\Omega A \nabla u \cdot n \, v = -\int_{\Gamma_N} \underbrace{A \nabla u \cdot n}_{= g_N} v = -\int_{\Gamma_N} g_N v$$

→ (Neuron cond. imposed in W. form – "natural" b.c.)

⇒ WP for the mixed problem:

Find $u \in V_{g_D}$:

$$a(u, v) = \int_\Omega f v + \int_{\Gamma_N} g_N v \qquad \forall v \in V_0$$

Note: assume $\exists u_g \in H^1(\Omega) : u_g|_{\Gamma_D} = g_D$

then $u = u_0 + u_g$; $u_0 \in V_0$

$a(u, v) = a(u_0, v) + a(u_g, v)$

□

⇑

Find $u_0 \in V_0$:

$$a(u_0, v) = \int_\Omega fv + \int_{\Gamma_H} g_H v - a(u_g, v) \quad \forall v \in V_0$$

The method of Galerkin (Q chap. 4)

Idea: restrict from V the problem into $V_h \subset V_0$

Consider case $(V, (\cdot,\cdot))$ Hilbert

assume $a : V \times V \to \mathbb{R}$ bilinear, continuous, coercive, $F \in V'$ (WP)

By LM $\exists!\ u \in V$: $a(u, v) = F(v)$ $\forall v \in V$

and $\|u\|_V \leq \alpha \|F\|_{V'}$

Galerkin method:

① Let $\{V_h\}_h$ a sequence of subspaces of V

(2) Restrict: find $u_h \in V_h$:

$$a(u_h, v_h) = F(v_h) \quad \forall v_h \in V_h$$

By Lax-Milgram $\forall h$ the Galerkin method is well-posed,

$$\|u_h\|_V \leq \alpha \|F\|_{V'} \quad \text{(Stability)}$$

Theorem (Galerkin orthogonality):

$$a(u - u_h, v_h) = 0 \quad \forall v_h \in V_h$$

Proof: $\begin{cases} a(u, v_h) = F(v_h) & \text{(WP)} \\ a(u_h, v_h) = F(v_h) & \text{(GM)} \end{cases} \quad \forall v_h \in V_h$

Subract to get the result. ∎

Note $a(u, v_h) - F(v_h) = 0$ Full (consistency)

Lemma (Céa): Under the LM assumptions

$$\|u - u_h\|_V \leq \left(\frac{\gamma}{\alpha}\right) \inf_{v_h \in V_h} \|u - v_h\|_V$$

(quasi-optimality)

Proof:

$$\alpha \|u - u_h\|_V^2 \leq a(u - u_h, u - u_h) \quad \text{(coercivity of } a\text{)}$$

$$= a(u - u_h, u - v_h) \quad \text{(Gal. orth.)}$$

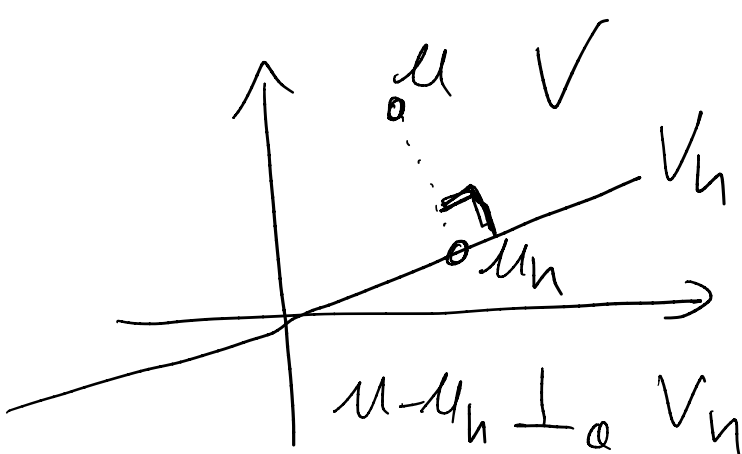
$$\leq \gamma \|u - u_h\|_V \|u - v_h\|_V \quad \text{(cont. of } a\text{)}$$

$$\|u - u_h\|_V \leq \frac{\gamma}{\alpha} \|u - v_h\|_V \quad \forall v_h \in V_h$$

Lemma (Céa symmetric case). If also a is symmetric,

(1) u_h is the orthogonal proj of u into V_h w.r.t. to inner product induced by a.

($a(w,v) = (w,v)_a$ is indeed an inner product if a is coercive and symmetric)



(2) $\|u - u_h\|_V \le \sqrt{\frac{\gamma}{\alpha}} \min_{v_h \in V_h} \|u - v_h\|_V$

$\|u - u_h\|_a = \min_{v_h \in V_h} \|u - v_h\|_a$

induced from $(\cdot,\cdot)_a$

Proof: Galerkin orthogonality

$= a(u - u_h, v_h) = 0 \quad \forall v_h \in V_h$

$= (u - u_h, v_h)_a$

that is $u - u_h \perp V_h$

$\Rightarrow \|u - u_h\|_a = \min_{v_h \in V_h} \|u - v_h\|_a$

$$
\begin{aligned}
\text{So } \alpha \|u-u_h\|_a^2 &= \min_{v_h \in V_h} \|u-v_h\|_a \\
&= \min_{v_h} a(u-v_h, u-v_h) \\
&\leq \gamma \min_{v_h} \|u-v_h\|_V^2 \\
&= a(u-u_h, u-u_h) \\
&\geq \alpha \|u-u_h\|_V^2
\end{aligned}
$$

□

Convergence of Galerkin

Céa (stab + cons) implies convergence of

$$\lim_{h \to 0} \inf_{v_h \in V_h} \|v - v_h\|_V = 0 \quad \forall v \in V$$

Example ("theoretical method")

Suppose V separable, and

$\{\varphi_j\}_j$ complete orthonormal system

Define $V_n := \text{span}\{\varphi_1, \dots, \varphi_n\}$

then $\forall u \in V$, $\forall \epsilon > 0$, $\exists n$:

$\exists\, w_n \in V_n : \|u - w_n\| < \epsilon$

$\Rightarrow$ Galerkin using sequence

V_n is convergent

FEM: Galerkin method based on piecewise polynomials

FEM in 1D

Model problem ← diff. reac.

$$\mathcal{L}u := \begin{cases} -(Ku')' + cu = f & \Omega = (0,1) \\ u(0) = u(1) \end{cases}$$

$f \in L^2(0,1)$, $K(x) \geq K_0 > 0$, $c \in L^\infty(\Omega)$, $c \geq 0$

(WP): Find $u \in V = H_0^1(\Omega)$:

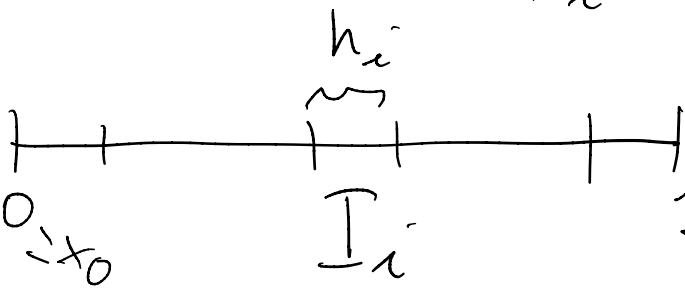
$$\underbrace{\int_0^1 Ku'v'\,dx + \int_0^1 cuv\,dx}_{a(u,v)} = \int_0^1 fv \qquad \forall v \in V$$

FEM of any order (k)

1) Fix partition $x_0 = 0$

$x_i = x_{i-1} + h_i$

$i = 1, \dots, N$



$I_i = [x_{i-1}, x_i)$

$h = \max_i h_i$

2) $V_h^k \subset V = H_0^1(\Omega)$

$$V_h^k := \left\{ v \in C_0(\bar{\Omega}) : v \in \mathbb{P}^k(I_i) \ \forall i \right\}$$

C_0: cont. and zero at the end points

($k = 1 \Rightarrow$ back to linear FEM)

3) The FEM: Find $u_h \in V_h^k$:

$$a(u_h, v_h) = \int_0^1 f v_h \quad \forall v_h \in V_h$$

Algebric form of the FEM

- fix a basis $\{\varphi_j\}_j$ of V_h^k
- write $u_h = \sum_j U_j \varphi_j$
- Test the FEM with $\varphi_j \ \forall j$

$$\sum_j U_j \left[\int_0^1 \kappa \varphi_j' \varphi_m' \, dx + \int_0^1 c \varphi_j \varphi_m \, dx \right] = \int_0^1 f \varphi_m$$

S_{mj}

$\forall m = 1, \dots, \dim V_h^k$

M_{mj} F_m

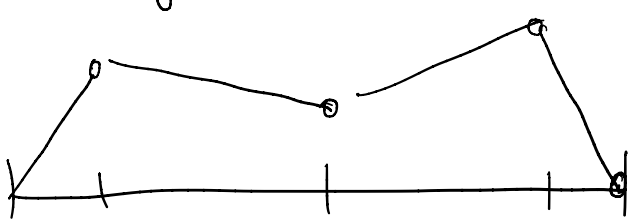
stiffness = S, mass = M

$U = (U_i)$

$$\begin{cases} AU = F \\ A = S + M \end{cases}$$

... depending on fixing the basis

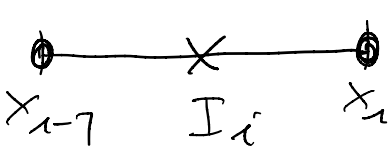
(1) $k = 1$



Lagrange basis $\varphi_j(x_i) = \delta_{ij}$

$\Rightarrow$ # DoF (unknowns) = # internal nodes

(2) $k = 2$ $\quad \leftarrow P_2(I_i)$



fix local (to the interval) quadratic function by imposing value at

- end points
- mid point

(3) k general I_i

on each interval, need to fix a k-degree polynomial

- end points
- k-1 distinct point inside I_i

$$\dim V_h^k = (N-1) + N \cdot (k-1)$$

A priori analysis of FEM

Céa: $\|u-u_h\|_V \leq \frac{\gamma}{\alpha} \inf_{v_h \in V_h^k} \|u-v_h\|_V$

($V = H_0^1$)

for some "special" $v_h = I_h u \in V_h^k$

$$\inf_{v_h} \|u-v_h\|_V \leq \|u - I_h u\|_V$$

$$I_h u(x_j^i) = u(x_j^i)$$

(local interpolant). Then

$$I_h u : V \longrightarrow V_h \quad \text{by}$$

$$I_h u|_{I_i} = \text{local interpolant}$$

(LT)

Theorem (Interpolation error estimate): $\forall v \in V$

$$\|v - I_h v\|_{L^2(I_i)} \leq C h_i^{k+1} |v|_{H^{k+1}(I_i)}$$

$$|v - I_h v|_{H^1(I_i)} \leq C h_i^{k} |v|_{H^{k+1}(I_i)}$$

from this, also, we have the global estimates

$$\|v - I_h v\|_{L^2(\Omega)} \leq C h^{k+1} |v|_{H^{k+1}(\Omega)}$$

$$(*)\quad |v - I_h v|_{H^1(\Omega)} \le C_I h^k |v|_{H^{k+1}(\Omega)}$$

Using $(*)$ on Lea gives

Theorem (Convergence of FEM).

If $u \in H^{k+1}(\Omega)$ then

$$|u - u_h|_{H^1(\Omega)} \le C(\alpha, \gamma, C_I)\, h^k |v|_{H^{k+1}(\Omega)}$$

FEM with degree k is of order k in H^1 norm